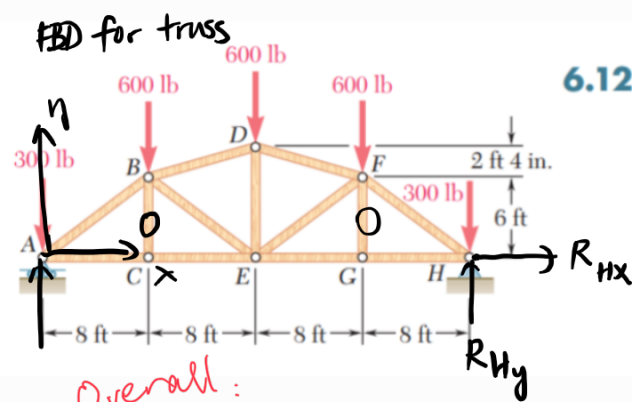


FBD for truss

Grade: 5/5

6.12 Determine the force in each member of the Gambrel roof truss shown. State whether each member is in tension or compression.



$$\sin \theta = \frac{6}{10}$$

$$\cos \theta = \frac{8}{10}$$

Overall:

$$\sum F_x = 0 \Rightarrow R_{Hx} = 0 \quad \checkmark$$

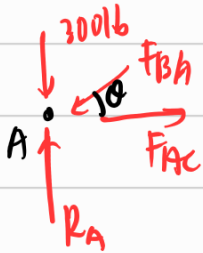
$$\sum F_y = 0 \quad \checkmark \quad -300 - 600 - 600 - 600 - 300 + R_A + R_{Hy} = 0$$

$$R_A + R_{Hy} = 2400 \text{ lb}$$

By symmetry, $R_A = R_{Hy} = 1200 \text{ lb}$

You need to prove it.
This can be obtained by doing moment balance.

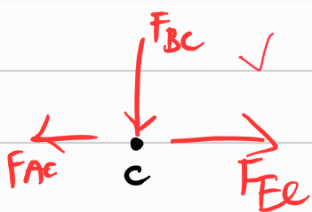
At A, $\checkmark$



$$\sum F_y = 0 \Rightarrow F_{AB} \sin \theta + 300 = 1200 \text{ lb} \quad F_{AB} = 1500 \text{ lb} \rightarrow C$$

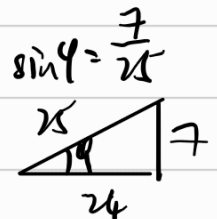
$$\sum F_x = 0 \Rightarrow F_{AB} \cos \theta = F_{AC} \quad F_{AC} = 1500 \left(\frac{8}{10} \right) = 1200 \text{ lb} \rightarrow T$$

At C, $\checkmark$



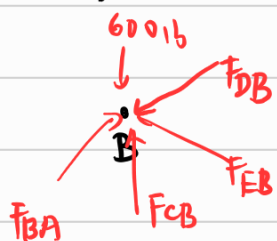
$$\sum F_y = 0 \Rightarrow F_{BC} = 0$$

$$\sum F_x = 0 \Rightarrow F_{AC} = F_{EC} = 1200 \text{ lb} \rightarrow T$$



$$\cos \phi = \frac{24}{25}$$

At B, $\checkmark$



$$\sum F_y = 0 \Rightarrow F_{AB} \sin \theta - 600 + F_{EB} \sin \theta + F_{CB} - F_{DB} \sin \phi = 0$$

$$900 - 600 + \frac{3}{5} F_{EB} + 0 - \frac{7}{25} F_{DB} = 0$$

$$\frac{3}{5} F_{EB} - \frac{7}{25} F_{DB} = -300$$

$$F_{EB} = \frac{7}{15} F_{DB} - 500 = \frac{7}{15} (1200) - 500 = 60 \text{ lb} \rightarrow C$$

$$\sum F_x = 0 \Rightarrow F_{BA} \cos \theta - F_{DB} \cos \phi - F_{EB} \cos \theta = 0$$

$$1200 - \frac{24}{25} F_{DB} - \frac{4}{5} F_{EB} = 0$$

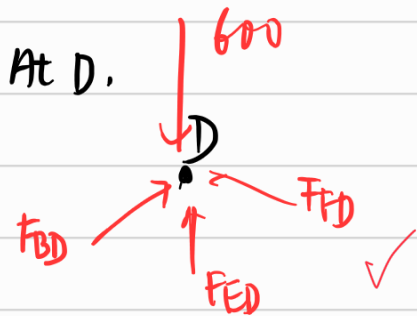
$$20 F_{EB} = 30000 - 24 F_{DB}$$

$$20\left(\frac{2}{3}F_{DB} - 500\right) = 30000 - 24F_{DB}$$

$$\frac{28}{3}F_{DB} - 10000 = 30000 - 24F_{DB}$$

$$\frac{100}{3}F_{DB} = 40000$$

$$F_{DB} = 1200\text{ lb} \rightarrow C$$



$$\sum F_x = 0 \Rightarrow F_{BD} \cos \varphi - F_{FD} \cos \varphi = 0$$

$$F_{BD} = F_{DB} = F_{FD} = 1200\text{ lb} \rightarrow C$$

$$\sum F_y = 0 \Rightarrow F_{BD} \sin \varphi + F_{FD} \sin \varphi - 600 + F_{ED} = 0$$

$$F_{ED} = 72\text{ lb} \rightarrow T$$

By symmetry,

$$F_{EF} = F_{BE} = 60\text{ lb}$$

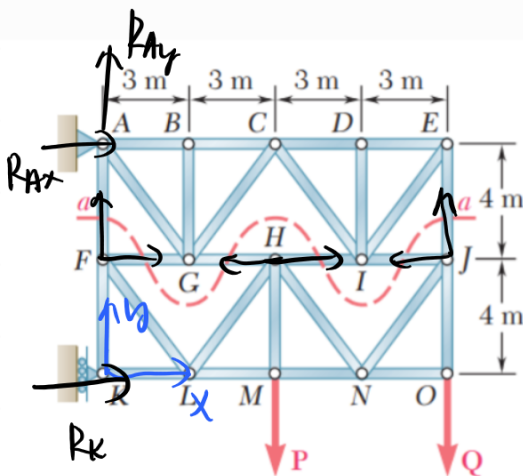
$$F_{EG} = F_{EC} = 1200\text{ lb}$$

$$F_{FG} = F_{BC} = 0$$

$$F_{AE} = F_{HG} = 1200\text{ lb}$$

$$F_{FH} = F_{BA} = 1500\text{ lb}$$

compression $\Rightarrow F_{BA} = F_{FH} = 1500\text{ lb}$, $F_{BE} = F_{EF} = 60\text{ lb}$, $F_{BD} = F_{FD} = 1200\text{ lb}$
 tension $\Rightarrow F_{AC} = F_{HG} = 1200\text{ lb}$, $F_{EC} = F_{EG} = 1200\text{ lb}$, $F_{DE} = 72\text{ lb}$



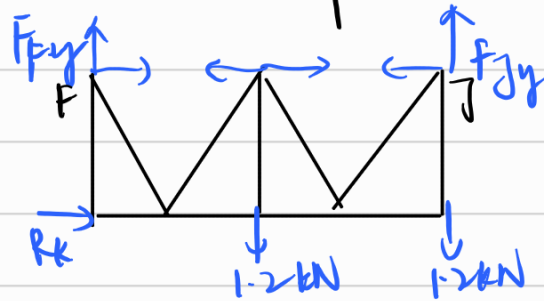
6.61 Determine the force in members AF and EJ of the truss shown when $P = Q = 1.2\text{ kN}$. (Hint: Use section aa .)

$$\sum \uparrow M_A = 0$$

$$R_K(8) - 1200(6) - 1200(12) = 0$$

$$R_K = 2700\text{ N}$$

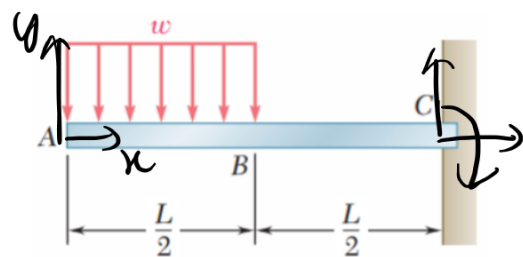
Consider lower part as a body,



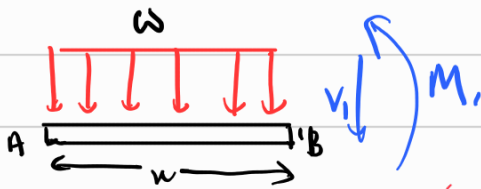
✓ Good labelling!

$$\begin{aligned} \sum M_F = 0 \quad & F_{Jy}(12) + R_k(4) - 1.2(6) - 1.2(12) = 0 \\ & F_{Jy} = 900 \text{ N}, T \end{aligned}$$

$$\begin{aligned} \sum M_D = 0 \quad & -F_{Jy}(12) + R_k(4) + 1.2(6) = 0 \\ & F_{Jy} = 1500 \text{ N}, T \end{aligned}$$



7.32 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.



$$\sum F_y = 0 \quad -wx - V_1 = 0 \quad V_1 = -\frac{wL}{2} \quad \text{at } x = \frac{L}{2}$$

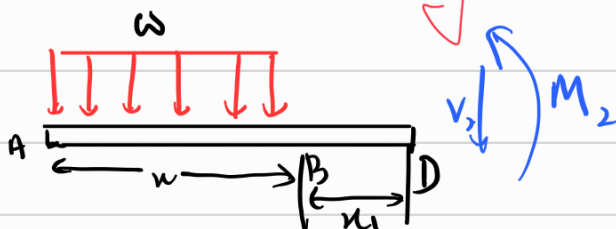
$$V_1 = -wx$$

$$M_1 = -\frac{1}{2}wx^2$$

$$+\sum M_B = 0 \quad M_1 + wx\left(\frac{x}{2}\right) = 0 \quad = -\frac{1}{2}w\left(\frac{L}{2}\right)^2$$

$$M_1 + \frac{wx^2}{2} = 0$$

$$= -\frac{1}{8}wL^2 \quad \text{at } x = \frac{L}{2}$$



$$V_2 = V_1 = -\frac{wL}{2} \quad \text{at } x = \frac{L}{2}$$

$$+\sum M_D = 0 \quad M_2 + wx\left(x_1 + \frac{x}{2}\right) = 0$$

$$M_2 = -\frac{wL}{2}\left(x_1 + \frac{L}{4}\right)$$

$$M_2 = -\frac{3}{8}wL^2 \quad \text{at } x_1 = \frac{L}{2}$$

